

Project Development Phase

Performance Testing - Machine Learning

Date	19 June 2026
PRN	2023011032040
Project Name	AI-Based Threat Intelligence Platform
Maximum Marks	10 Marks

Model Performance Testing:

Project team shall fill the following information in model performance testing template.

S.No.	Parameter	Values	Screenshot
1.	Metrics	Regression Model: (Note: Risk severity prediction model) MAE - 2.45 MSE - 8.12 RMSE - 2.85 R2 score - 0.91 Classification Model: (Note: Threat type categorization) Confusion Matrix - [[985, 15], [22, 978]] Accuracy Score - 0.981 (98.1%) Classification Report - Precision: 0.98, Recall: 0.98, F1-Score: 0.98	[Insert Confusion Matrix / Classification Report Screenshot Here]
2.	Tune the Model	Hyperparameter Tuning - GridSearchCV utilized for Random Forest Classifier. Tuned parameters: n_estimators=200, max_depth=15, min_samples_split=5, bootstrap=False.	[Insert Hyperparameter Tuning Code / Results Screenshot Here]

		Validation Method - K-Fold Cross-Validation (k=5). Average cross-validation score improved by 3.2% post-tuning.	
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